

Class 11 Physics: Motion in a Straight Line Formula Sheet

1. Average Velocity and Average Speed

Average Velocity is defined as the change in position or displacement (Δx) divided by the time intervals (Δt):

$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

Average Speed is defined as the total path length travelled divided by the total time interval:

$$\text{Average Speed} = \frac{\text{Total Path Length}}{\text{Total Time Interval}}$$

2. Instantaneous Velocity and Acceleration

Instantaneous Velocity (v) is the velocity at a specific instant of time:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Instantaneous Acceleration (a) is the rate of change of velocity with respect to time:

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Acceleration can also be expressed as:

$$a = v \frac{dv}{dx}$$

3. Kinematic Equations for Uniformly Accelerated Motion

For an object moving with constant acceleration a , initial velocity u , final velocity v , and displacement s in time t :

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$
- $s = \left(\frac{u+v}{2}\right)t$
- Displacement in the n^{th} second: $s_n = u + \frac{a}{2}(2n - 1)$

4. Motion Under Gravity (Free Fall)

Taking the upward direction as positive, the acceleration due to gravity is $a = -g$ (where $g \approx 9.8 \text{ m/s}^2$):

- $v = u - gt$
- $h = ut - \frac{1}{2}gt^2$
- $v^2 = u^2 - 2gh$

For an object dropped from rest ($u = 0$):

- Time to reach the ground: $t = \sqrt{\frac{2h}{g}}$
- Velocity on reaching the ground: $v = \sqrt{2gh}$

5. Relative Velocity

Relative velocity of object A with respect to object B :

$$v_{AB} = v_A - v_B$$

Relative velocity of object B with respect to object A :

$$v_{BA} = v_B - v_A$$

6. Important Graphical Relations

- The slope of the Position-Time ($x - t$) graph represents the velocity.
- The slope of the Velocity-Time ($v - t$) graph represents the acceleration.
- The area under the Velocity-Time ($v - t$) graph represents the displacement.
- The area under the Acceleration-Time ($a - t$) graph represents the change in velocity.

7. Stopping Distance and Reaction Time

Stopping distance (d_s) of a vehicle:

$$d_s = \frac{-u^2}{2a}$$

Reaction time (t_r) calculated from a ruler drop test:

$$t_r = \sqrt{\frac{2h}{g}}$$